### 21.1.2 Proteins

Proteins are the essential nutrients and also the building blocks of the body. They are essential for growth and repair of body cells and tissues. Proteins are made of amino acids.

Essential amino acids are those that cannot be biosynthesized by the body and must be obtained from the diet. The nine essential amino acids are phenylalanine, valine, threonine, tryptophan, methionine, leucine, isoleucine, lysine and histidine.

### 21.1.3 Fats

Fat in the diet provides energy. They maintain cell structures and are involved in metabolic functions.

Essential fatty acids cannot be synthesized in the body and are provided through diet. Essential fatty acids required in human nutrition are omega fatty acids.

### 21.1.4 Vitamins

Vitamins are the vital nutrients, required in minute quantities to perform specific physiological and biochemical functions.




**More to Know**

Dr. Funk introduced the term vitamin. Vitamin A was given the first letter of the alphabet, as it was the first vitamin discovered.




Human skin can synthesize Vitamin D when exposed to sunlight (especially early morning). When the sun rays falls on the skin dehydrocholesterol is converted into Vitamin D. Hence, Vitamin D is called as **Sunshine vitamin.** Vitamin D improves bone strength by helping body to absorb calcium.

### 21.1.5 Minerals

Minerals are inorganic substances required as an essential nutrient by organisms to perform various biological functions necessary for life. They are the constituents of teeth, bones, tissues, blood, muscle and nerve cells.

The **macrominerals** required by the human body are calcium, phosphorus, potassium, sodium and magnesium. The **microminerals** required by the human body also called **trace elements** are sulfur, iron, chlorine, cobalt, copper, zinc, manganese, molybdenum, iodine and selenium.

## 21.2 Protein Energy Malnutrition (PEM)

Absence of certain nutrients in our daily diet over a long period of time leads to deficiency diseases. This condition is referred as Malnutrition. Deficiency of proteins and energy leads to severe conditions like: Kwashiorkar and Marasmus.

**Table 21.1** Dietary sources of major foodstuffs

| Major food stuffs | Dietary sources | Daily requirements (grams) |
|---|---|---|
| Carbohydrates | Honey, sugarcane, fruits, whole grains, starchy vegetables, rice | 150-200 |
| Proteins | Legumes, pulses, nuts, soya bean, green leafy vegetables, fish, poultry products, egg, milk and dairy products | 40 |
| Fats | Egg yolk, saturated oil, meat | 35 |



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